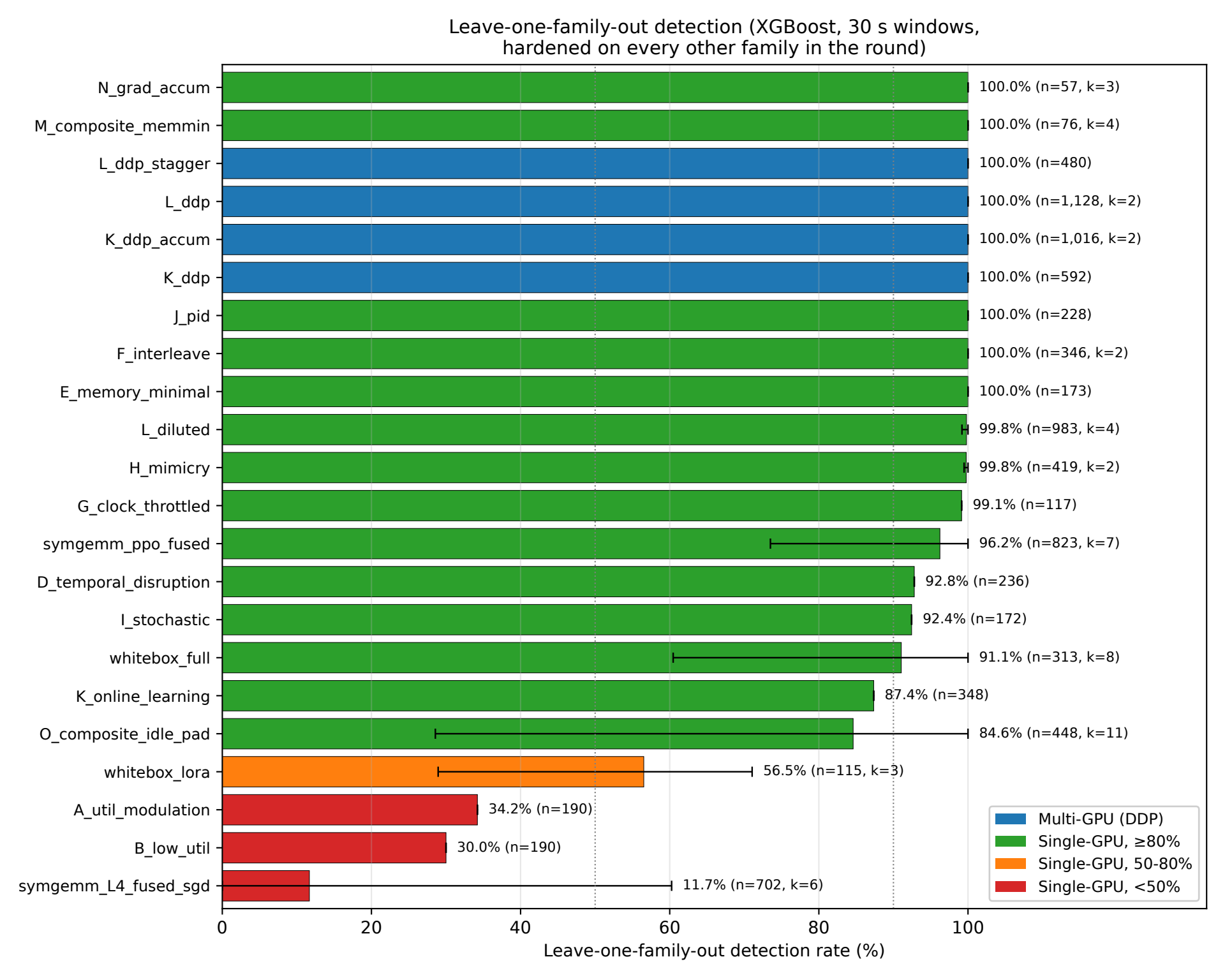
Leave-one-family-out detection (XGBoost, 30 s windows,
hardened on every other family in the round)
N_grad_accum
M_composite_memmin
L_ddp_stagger
L_ddp
K_ddp_accum
K_ddp
J_pid
F_interleave
E_memory_minimal
L_diluted
H_mimicry
G_clock_throttled
symgemm_ppo_fused
D_temporal_disruption
I_stochastic
whitebox_full
K_online_learning
O_composite_idle_pad
whitebox_lora
A_util_modulation
B_low_util
symgemm_L4_fused_sgd
100.0% (n=57, k=3)
100.0% (n=76, k=4)
100.0% (n=480)
100.0% (n=1,128, k=2)
100.0% (n=1,016, k=2)
100.0% (n=592)
100.0% (n=228)
100.0% (n=346, k=2)
100.0% (n=173)
99.8% (n=983, k=4)
99.8% (n=419, k=2)
99.1% (n=117)
96.2% (n=823, k=7)
92.8% (n=236)
92.4% (n=172)
91.1% (n=313, k=8)
87.4% (n=348)
84.6% (n=448, k=11)
56.5% (n=115, k=3)
34.2% (n=190)
30.0% (n=190)
11.7% (n=702, k=6)
Multi-GPU (DDP)
Single-GPU, ≥80%
Single-GPU, 50-80%
Single-GPU, <50%
0
20
40
60
80
100
Leave-one-family-out detection rate (%)